C PROGRAM MINI PROJECT

NAME: PRIYA DHARSINI M

REG NO:921724115033

DEPT: COMPUTER SCIENCE AND DESIGN

TRAINER NAME: BALAJI

SUBMISSION DATE: 26/02/2026

TITLE: HOSTEL MANAGEMENT SYSTEM

2.ABSTRACT:

Hostel Management System is a software application developed using C language to manage hostel student records efficiently. It helps in storing, updating, searching, and displaying student details like name, room number, department, and fees. The system reduces manual work and maintains accurate hostel records.

3.INTRODUCTION:

Hostel Management System is designed to manage the activities of a hostel. In many colleges, hostel data is maintained manually, which is time-consuming and error-prone. This system automates student record management using basic file handling concepts in C.

The system performs:

- Add student record
- Display student record
- Search student record
- Delete student record

4.OBJECTIVES:

- To maintain hostel student records digitally
- To reduce manual paperwork
- To provide easy access to student information
- To store and retrieve data using file handling in C
- To improve data accuracy

5.TOOLS AND TECHNOLOGIES :

- Programming Language: C
- Compiler: GCC / Turbo C / CodeBlocks
- Platform: Windows / Linux
- Concepts Used:
    - Structures
    - Functions
    - File Handling
    - Conditional Statements
    - Random Number Generation

6.SYSTEM REQUIREMENT:

Hardware:

- 4GB RAM minimum
- Intel i3 or above

Software:

- C Compiler (GCC)
- Code Editor (VS Code / CodeBlocks)

7.ALGORITHM:

1. Start
2. Display menu
3. Read user choice
4. If choice = 1 → Add student record
5. If choice = 2 → Display records
6. If choice = 3 → Search record
7. If choice = 4 → Exit
8. Stop

8.FLOWCHART:

```
   Start
     |
  Display Menu
     |
  Read Choice
 /   |    \
```

```
Add   Display   Search
  \    |     /
      Exit
       |
      Stop
```

9.PROGRAM CODE:

```
#include <stdio.h>

#include <stdlib.h>


struct student {

   int id;

   char name[50];

   int room_no;

   char dept[30];

};


void addStudent() {

   struct student s;

   FILE *fp;


   fp = fopen("hostel.txt", "a");


   printf("Enter Student ID: ");

   scanf("%d", &s.id);
```

```
    printf("Enter Name: ");
    scanf("%s", s.name);
    printf("Enter Room Number: ");
    scanf("%d", &s.room_no);
    printf("Enter Department: ");
    scanf("%s", s.dept);

    fwrite(&s, sizeof(s), 1, fp);
    fclose(fp);

    printf("Record Added Successfully!\n");
}

void displayStudent() {
    struct student s;
    FILE *fp;

    fp = fopen("hostel.txt", "r");

    while(fread(&s, sizeof(s), 1, fp)) {
        printf("\nID: %d", s.id);
        printf("\nName: %s", s.name);
        printf("\nRoom No: %d", s.room_no);
        printf("\nDepartment: %s\n", s.dept);
    }
```

```
    fclose(fp);
}

int main() {
    int choice;

    while(1) {
        printf("\n--- Hostel Management System ---");
        printf("\n1. Add Student");
        printf("\n2. Display Students");
        printf("\n3. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                addStudent();
                break;
            case 2:
                displayStudent();
                break;
            case 3:
                exit(0);
            default:
```

```
            printf("Invalid Choice!");
        }
    }

    return 0;
}
```

10.OUTPUT:

```
--- Hostel Management System ---
1. Add Student
2. Display Students
3. Exit
Enter your choice: 1

Enter Student ID: 101
Enter Name: Abinaya
Enter Room Number: 12
Enter Department: CSE

Record Added Successfully!
```

ID: 101

NAME: JANANI

ROOM NO: 12

DEPARTMENT: CSE

11.RESULT:

- Thus the above program executed successfully.

12.APPLICATION:

- College Hostels
- School Hostels
- Paying Guest (PG) Management
- Residential Training Centers

13.CONCLUSION:

The **Hostel Management System** developed using C language successfully manages hostel student records in a simple and efficient manner. The system performs basic operations such as adding and displaying student details using file handling concepts.

It reduces manual work, minimizes errors, and ensures proper storage of student information. By using structures, functions, and file operations in C, the project demonstrates practical implementation of core programming concepts.

Thus, the project achieves its objective of creating a simple, reliable, and user-friendly hostel record management system.

14.FUTURE ENHANCEMENT:

- Add fee payment tracking
- Add delete and update options
- Add password security
- Convert into GUI application
- Connect with database (MySQL)

15.REFERENACE:

 . Balagurusamy – Programming in ANSI C

 C Programming Notes

 www.geeksforgeeks.org

 College C Programming Lab Manual